



Research Innovation and Technical Writing CSE414 (Lec.-5)

Qualitative & Quantitative
Techniques; Ethical Data Handling

Privacy, Consent, and Integrity
Research Methodology & Innovation

Objectives

- ▶ To differentiate qualitative and quantitative techniques
- ▶ To select appropriate data collection methods
- ▶ To understand privacy, consent, and integrity
- ▶ To apply ethical data handling practices

Why Data Collection Matters?

- ▶ Supports research questions
- ▶ Tests hypotheses
- ▶ Enables evidence-based decisions

Research Techniques Overview

- ▶ **Qualitative:** non-numerical, exploratory
- ▶ **Quantitative:** numerical, statistical
- ▶ **Mixed methods** combine both approaches

What is Qualitative Research?

- ▶ Focuses on experiences, opinions, and behaviors
- ▶ Uses interviews, observations, focus groups



Qualitative Data Collection Methods

- ▶ Interviews
- ▶ Focus Groups
- ▶ Observations
- ▶ Case Studies
- ▶ Document Analysis

Advantages and Limitations of Qualitative Research

- ▶ Advantages: rich insights, flexibility
- ▶ Limitations: time-consuming, difficult to generalize

What is Quantitative Research?

- ▶ Focuses on numerical data and measurement
- ▶ Supports statistical analysis and comparisons



Quantitative Data Collection Methods

- ▶ Surveys
- ▶ Questionnaires
- ▶ Experiments
- ▶ Sensors
- ▶ Databases

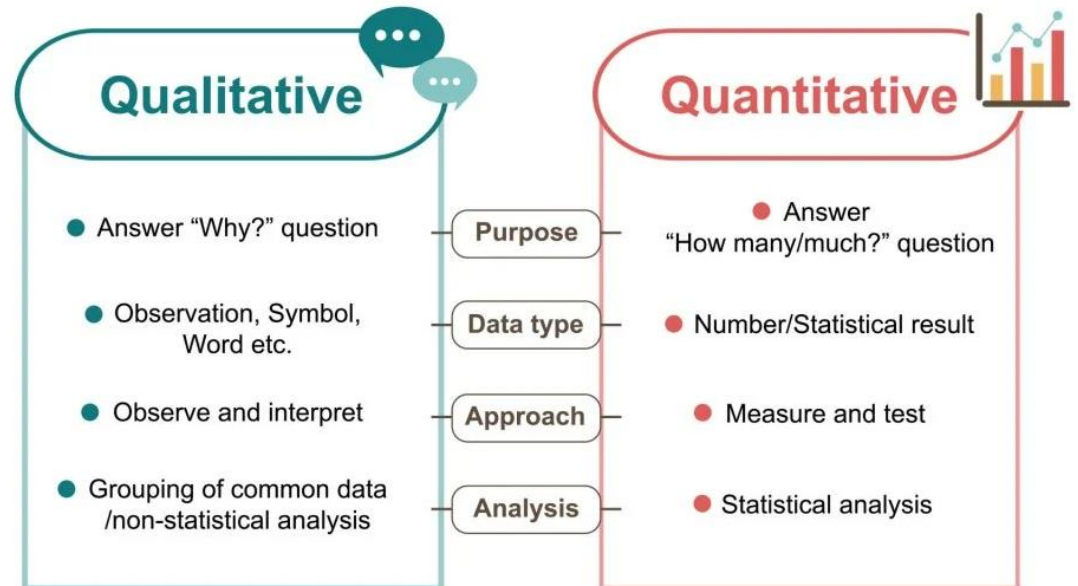
Advantages and Limitations of Quantitative Research

- ▶ Advantages: objective, scalable
- ▶ Limitations: limited depth, depends on data quality

Qualitative vs Quantitative Comparison

- ▶ Text vs Numbers
- ▶ Small vs Large samples
- ▶ Themes vs Statistics

Type of research design



Mixed Methods Research

- ▶ Combines qualitative and quantitative approaches
- ▶ Provides comprehensive findings

Introduction to Research Ethics

- ▶ Protect participants
- ▶ Ensure trustworthy results
- ▶ Maintain public confidence

8 Key Research Ethics Committee Guidelines



Participant
Protection



Risk-Benefit
Balance



Informed
Consent



Vulnerable
Populations



Scientific
Validity



Researcher
Qualifications



Data
Protection



Participant
Information

Ethical Data Handling

- ▶ Privacy
- ▶ Consent
- ▶ Integrity

Privacy in Research

- ▶ Protect personal information
- ▶ Use anonymization, encryption, and access control

Informed Consent

- ▶ Voluntary participation
- ▶ Participants understand risks and benefits
- ▶ Documented agreement

Example of Consent Process

- ▶ Explain purpose
- ▶ Ensure confidentiality
- ▶ Obtain consent
- ▶ Allow withdrawal

Data Integrity

- ▶ Maintain accuracy, consistency, and reliability
- ▶ Avoid fabrication and falsification

Ethical Violations and Consequences

- ▶ Plagiarism
- ▶ Privacy breaches
- ▶ Loss of credibility
- ▶ Legal penalties



Case Study Discussion

- ▶ Sharing participant data without permission
- ▶ Identify ethical issues and solutions

Best Practices for Ethical Data Handling

- ▶ Obtain consent
- ▶ Protect privacy
- ▶ Store data securely
- ▶ Report findings honestly

Summary

- ▶ Qualitative and quantitative techniques
- ▶ Privacy, consent, and integrity are essential
- ▶ Ethics improves trust and research quality

Thank You